$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = (A \oplus B) C_{in} + A \cdot B$$

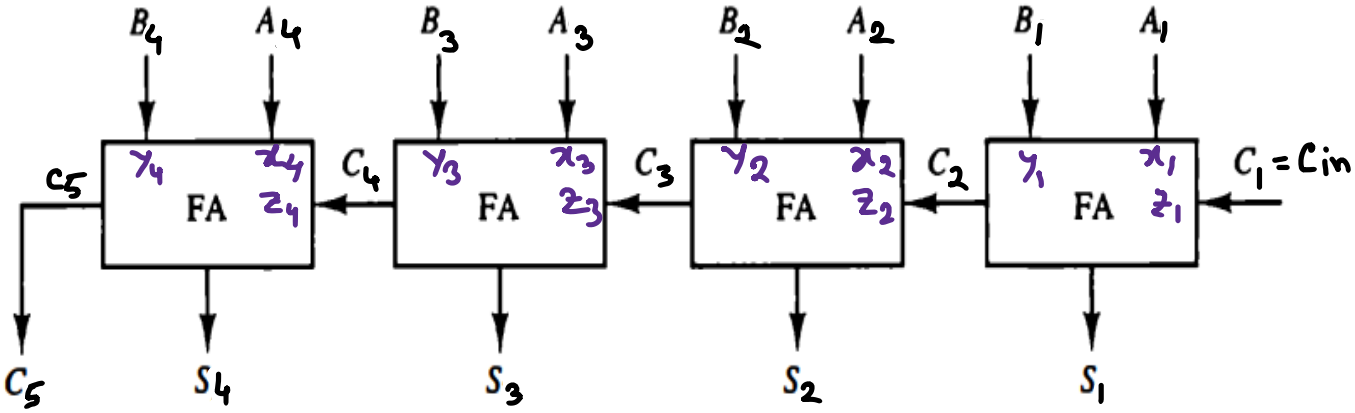


Figure 4-bit binary adder.

Function Table for the Arithmetic Circuit of Fig. 9-8 (Book: DLCD-Morris Mano)

Selection			Inputs			Output			
						Cout	Sum		
S1	**S0**	**Cin=C1**	**Xi**	**Yi**	**Z1=Cin=C1**	**Zi, i≠1**	**Output F =A + B + Cin = A ⊕ B ⊕ Cin**	**Y equals to (for 4 bits)**	**Functions**
0	0	0	**Ai**	**0**	**0**	**Ci**	F=Ai+0+0 =Ai	**0**	Transfer of A
0	0	1	**Ai**	**0**	**1**	**Ci**	F=Ai+0+1 = Ai+1	**0**	Increment of A
0	1	0	**Ai**	**Bi**	**0**	**Ci**	F=Ai+Bi+0 = Ai+Bi	**Bi**	Addition of A and B
0	1	1	**Ai**	**Bi**	**1**	**Ci**	F=Ai+Bi+1	**Bi**	Addition with Carry = **A+B+1**
1	0	0	**Ai**	**Bi'**	**0**	**Ci**	F=Ai+Bi'+0 = Ai-Bi-1 Bi'+1=-Bi Bi'=-Bi-1	**Bi'**	Subtraction with Borrow = **A-B-1**
1	0	1	**Ai**	**Bi'**	**1**	**Ci**	F=Ai+Bi'+1 = Ai-Bi Bi'+1 = -Bi	**Bi'**	Subtraction of A and B
1	1	0	**Ai**	**1**	**0**	**Ci**	F=Ai-1	**1111 = -1 in 2's**	Decrement of A
1	1	1	**Ai**	**1**	**1**	**Ci**	F=Ai-1+1 = Ai	**1111= -1 in 2's**	Transfer of A

Xi = Ai

Yi = (S1'.S0.Cin' + S1'.S0.Cin).Bi + (S1.S0'.Cin'+ S1.S0'.Cin).Bi' + (S1.S0.Cin'+S1.S0.Cin)

= S1'.S0.Bi.(Cin'+Cin) + S1.S0'.Bi'.(Cin'+Cin) + S1.S0.(Cin' +Cin)

= S1'.S0.Bi + S1.S0'.Bi'+ S1.S0

= S1'.S0.Bi + S1.S0'.Bi'+ S1.S0.(Bi+Bi')

= S1'.S0.Bi + S1.S0'.Bi'+ S1.S0.Bi + S1.S0.Bi'

= S0.Bi.(S1'+S1) + S1.Bi'.(S0'+S0)

= **S0.Bi + S1.Bi'**

Z1 = S1'.S0'.Cin + S1'.S0.Cin + S1.S0'.Cin + S1.S0.Cin

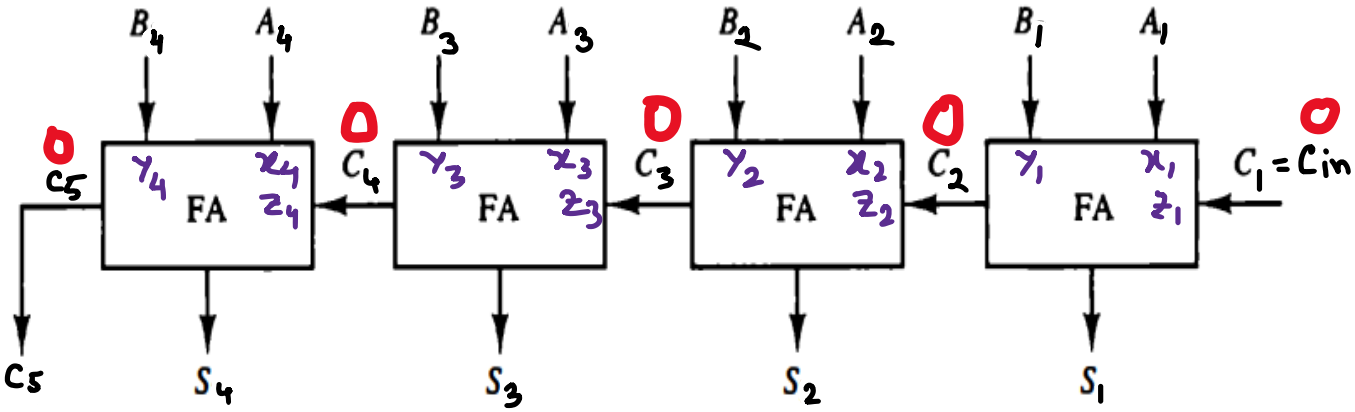
= Cin.(S1'.S0' + S1'.S0 + S1.S0' + S1.S0)

= Cin.(S1'.(S0'+S0) + S1(S0'+S0))

= Cin.(S1'+S1) = **Cin = C1**

Zi = Ci

Logic Unit using Full Adder



For Logic operations, Cin = C1 = C2 = C3 = C4 = C5 = 0

New Selection Variable = S2
When S2=0 -> Do arithmetic operations
When S2=1 -> Do logic operations

OR:
(Ai + Bi) ⊕ 0 ⊕ 0
= **(Ai + Bi)**

OR Gate, not addition

XOR:
Ai ⊕ Bi ⊕ 0
= **Ai ⊕ Bi**

AND:
(Ai+Bi') ⊕ Bi' ⊕ 0
= (Ai+Bi') ⊕ Bi'
= (Ai+Bi')'Bi' + (Ai+Bi')Bi
= Ai'.Bi.Bi' + Ai.Bi + Bi'.Bi
=Ai'.0 + Ai.Bi + Bi'.Bi
=**Ai.Bi**

$x \cdot x' = 0$ | $x + 0 = x$
$x \cdot 0 = 0$
$x \oplus 0 = x$
$x \oplus y = x'y + xy'$
$(x+y)' = x'y'$
$(x')' = x$

NOT:
Ai ⊕ 1 ⊕ 0 = Ai ⊕ 1 = Ai.1' + Ai'.1 = Ai.0 + Ai' = **Ai'**

OR Gate, not addition

XNOR:
Ai ⊕ Bi' ⊕ 0 = Ai ⊕ Bi' = Ai'.Bi' + Ai.Bi = **Ai xnor Bi**

Function Table for the ALU of Fig. 9-13 (Book: DLCD-Morris Mano)

Selection				Inputs			Output			
							Cout	Sum		
S2	**S1**	**S0**	**Cin=C1**	**Xi**	**Yi**	**Z1=Cin=C1**	**Zi, i≠1**	**Output F =A + B + Cin = A ⊕ B ⊕ Cin**	**Y equals to (for 4 bits)**	**Functions**
0	0	0	0	**Ai**	**0**	**0**	**Ci**	F=Ai+0+0 =Ai	**0**	Transfer of A
0	0	0	1	**Ai**	**0**	**1**	**Ci**	F=Ai+0+1 = Ai+1	**0**	Increment of A
0	0	1	0	**Ai**	**Bi**	**0**	**Ci**	F=Ai+Bi+0 = Ai+Bi	**Bi**	Addition of A and B
0	0	1	1	**Ai**	**Bi**	**1**	**Ci**	F=Ai+Bi+1	**Bi**	Addition with Carry = **A+B+1**
0	1	0	0	**Ai**	**Bi'**	**0**	**Ci**	F=Ai+Bi'+0 = Ai-Bi-1 Bi'+1=-Bi Bi'=-Bi-1	**Bi'**	Subtraction with Borrow = **A-B-1**
0	1	0	1	**Ai**	**Bi'**	**1**	**Ci**	F=Ai+Bi'+1 = Ai-Bi Bi'+1 = -Bi	**Bi'**	Subtraction of A and B
0	1	1	0	**Ai**	**1**	**0**	**Ci**	F=Ai-1	**1111 = -1 in 2's**	Decrement of A
0	1	1	1	**Ai**	**1**	**1**	**Ci**	F=Ai-1+1 = Ai	**1111= -1 in 2's**	Transfer of A
1	0	0	0	**Ai+Bi**	**0**	**0**	**0**	F=Ai + Bi	**All 0's**	OR
1	0	1	0	**Ai**	**Bi**	**0**	**0**	F=Ai ⊕ Bi	**Bi**	XOR
1	1	0	0	**Ai+Bi'**	**Bi'**	**0**	**0**	F=Ai.Bi	**Bi'**	AND
1	1	1	0	**Ai**	**1**	**0**	**0**	F=Ai⊕1⊕0 =Ai'	**All 1's, but now, It is not -1**	NOT

Xi = Ai + S2.S1'.S0'(Ai+Bi) + S2.S1.S0'(Ai+Bi')
= Ai + S2.S1'.S0'.Ai + S2.S1.S0'.Ai + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'
= Ai + S2.S0'Ai.(S1'+S1) + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'
= Ai + S2.S0'.Ai + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'
= Ai(1+S2.S0') + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'
= Ai.1 + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'
= **Ai + S2.S1'.S0'.Bi + S2.S1.S0'.Bi'**

S2.S0'(S1'.Bi + S1.Bi')
S2.S0'(S1 xor Bi)
Xi = Ai + S2.S0'(S1 xor Bi)

Yi = (S2'.S1'.S0.Cin' + S2'.S1'.S0.Cin + S2.S1'.S0).Bi +
(S2'.S1.S0'.Cin' + S2'.S1.S0'.Cin + S2.S1.S0').Bi' +
(S2'.S1.S0.Cin' + S2'.S1'.S0.Cin + S2.S1.S0).1

= (S2'.S1'.S0 + S2.S1'.S0). Bi + (S2'.S1.S0' + S2.S1.S0').Bi' + (S2'.S1.S0 + S2.S1.S0)
= S0.S1'.(S2' + S2).Bi + S1.S0'.(S2'+S2).Bi' + S1.S0.(S2'+S2)
= S0.S1'.Bi + S1.S0'. Bi' + S1.S0
= S0.S1'.Bi + S1.S0'. Bi' + S1.S0.(Bi+Bi')
= S0.S1'.Bi + S1.S0'. Bi' + S1.S0.Bi + S1.S0.Bi'
= S0.Bi(S1'+S1) + S1.Bi'(S0'+S0)
= **S0.Bi + S1.Bi'**

Z1 = S2'.S1'.S0'.Cin + S2'.S1'.S0.Cin + S2'.S1.S0'.Cin + S2'.S1.S0.Cin
= S2'.Cin.(S1'.S0' + S1'.S0 + S1.S0' + S1.S0)
= S2'.Cin.(S1'.(S0'+S0) + S1(S0'+S0))
= S2'.Cin.(S1'+S1)
= **S2'.Cin**

Zi = S2'.Ci